MAKHLUG JAFAROV

SOFTWARE ENGINEER | Padova, Italy

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PROFESSIONAL SUMMARY

Software Engineer with a background in C++ systems and Python backend development, increasingly focused on building AI-powered internal tools and GPT-based systems. Shipped production AI products including a RAG pipeline on GCP (Gemini + Vertex AI) and a Gemini-powered property management automation bot. Currently working on automotive simulation infrastructure at Huxelerate while studying MSc Computer Science at the University of Padova. Strong on backend architecture, structured system design, and turning logic frameworks into working, deployed tools.

WORK EXPERIENCE

Software Engineer — Jul 2025 – Present

Huxelerate - Milan, Italy (Remote)

- Engineered Python automation scripts to parse and analyze complex ARXML configurations for automotive simulation tools
- Implemented and optimized RESTful API endpoints using Python Flask to enhance core platform functionality
- Managed version control and code integration via Git and GitHub within a fully remote Agile development cycle

Mobility Support Specialist — Jan 2023 – Dec 2023

Sakon - United States (Remote)

- Managed multi-channel technical support for Fortune 500 clients
- Used Jira and TEM systems to track and coordinate technical issues with engineering teams

C++ Developer — Dec 2022 – Dec 2023

Ericsson - Budapest, Hungary (Hybrid)

- Modernized legacy C++ modules in MTAS Media Framework using modern STL functions and design patterns
- Performed troubleshooting using Wireshark for protocol analysis, including two-month investigation to fix stability issues
- Developed and documented automated function and system tests using Jenkins and internal Tiger tool
- Worked in Agile/Scrum environment using Jira, Gerrit, and Git for task management and code review

Backend Developer — Nov 2022 – Feb 2023

Audivisa GmbH - Germany (Remote)

- Developed backend infrastructure for DiGlu School Management System using PHP and Joomla framework
- Built database solutions with MySQL and implemented API endpoints
- Fixed bugs and maintained application stability

C++ Developer Intern — Mar 2022 – Dec 2022

Ericsson - Budapest, Hungary (Hybrid)

- Refactored legacy C++ code to modern C++11/14 standards
- Contributed to Agile sprint cycles using Jira and Gerrit
- Promoted to full-time C++ Developer role after 10 months

Expert Contributor — Mar 2021 – Mar 2022

Quizlet (Remote)

- Authored technical solutions for data structures, algorithms, and system design topics
- Formatted content using LaTeX and KaTeX

EDUCATION

MSc Computer Science — Oct 2025 – Aug 2027

University of Padova - Padova, Italy

Award: Regional Scholarship

BSc Computer Science — Sep 2021 – Jul 2025

Eötvös Loránd University - Budapest, Hungary

Award: Stipendium Hungaricum

Thesis: Servx (iOS App & Backend)

PROJECTS

DİM AI Exam Tutor
Python · TypeScript · React · GCP

AI-powered exam preparation platform targeting ~120,000 annual Azerbaijani university entrance exam candidates lacking access to affordable, curriculum-aligned tutoring.

- Built a full-stack RAG pipeline using Google Gemini 1.5 Flash and Vertex AI Vector Search, achieving 85%+ answer accuracy on official past exam questions
- Designed serverless GCP architecture with Cloud Functions (2nd Gen), Firestore, Pub/Sub, and Eventarc for an automated document ingestion and embedding pipeline
- Implemented multimodal OCR input via Google Document AI, enabling students to photograph textbook questions and receive step-by-step solutions with source citations
- Provisioned all cloud infrastructure with Terraform; built React/TypeScript frontend with Firebase Auth and multilingual support (Azerbaijani, Russian, English)

Property Management Operations Bot
Python · FastAPI · PostgreSQL · Gemini AI

Multi-platform property management automation bot digitizing daily operations for a short-term rental business across Airbnb and Booking.com.

- Integrated Slack and Gmail as data inputs; used Google Gemini AI to parse unstructured check-in and cleaning lists into structured PostgreSQL records
- Built a Telegram bot interface delivering real-time alerts, overbooking conflict resolution with interactive UI, and 20+ operational slash commands for field staff
- Implemented a two-stage automated reminder system and deployed on Render with a FastAPI backend

ImSolTech — EdTech Platform
Next.js 15 · TypeScript · PostgreSQL · Vitest

Full-stack EdTech platform enabling IT course enrollment, career assessment, and admin management.

- Designed an FSM-based chatbot with three operational modes (advisor, lead generation, registration) using a typed rule engine for structured, deterministic conversation flows
- Built a full admin dashboard with CRM, course/track CRUD, enrollment management, real-time SQL-backed notifications, and user request handling
- Implemented secure auth with bcrypt, email verification via Resend, rate limiting, and GDPR-compliant Google Analytics Consent Mode v2

GLDP.az — Education Consulting Platform
Next.js · TypeScript · PostgreSQL · Prisma

Full-stack, multilingual (AZ/EN/RU) education consulting platform for IELTS preparation and international study placements.

- Built with Next.js 15 (App Router) and React 19; implemented i18n routing with full SEO optimization across three languages
- Developed a secure role-based admin dashboard for managing course packages, scholarship programs, student registrations, and dynamic content
- Integrated Cloudinary for media management, Nodemailer for automated email handling, and a full testing suite using Vitest, Playwright, and React Testing Library

Servx — Service Marketplace App (iOS + Backend)
Swift · Java · Spring Boot · PostgreSQL

Dual-role service marketplace iOS app connecting service seekers with providers (plumbers, cleaners, electricians). Built independently from design to deployment.

- Developed iOS frontend in SwiftUI following strict MVVM architecture and a RESTful Spring Boot 3 backend with PostgreSQL
- Implemented real-time chat and push notifications via WebSocket/STOMP and a JWT-based auth system with email verification
- Engineered a multi-state booking workflow (Pending → Accepted → Confirmed → Completed) managing concurrency and state transitions across both client and server

SKILLS

Programming Languages & Core: Python, C++, Java / Spring Boot, Swift

Tools & Dev Compilers: Jira, Wireshark, Jenkins, LLVM, CMake, Git

Cloud & Containers: AWS, GCP, Terraform, Docker

AI & LLM Tooling: OpenAI API, Google Gemini API, Vertex AI, RAG, Agent-based Workflows, Prompt Engineering & GPT System Design, AI-Driven Development, Google AI Studio, Claude Code

Databases: PostgreSQL / MySQL, Firestore

SOFT SKILLS: Technical Communication, Problem Solving & Troubleshooting, Time Management & Prioritization

LANGUAGES: Azerbaijani, English, Turkish